SEARCH FUNNEL EXPERIMENT

SYNTHETIC PRODUCT ANALYTICS CASE STUDY

AI-assisted portfolio demonstration prepared for Rahul Kashyap. All records are generated; no Microsoft, Bing, employer, or customer data is used.

THE PRODUCT DECISION

Should a hypothetical search-results change roll out? Evaluate session conversion as the primary metric and mean latency as a guardrail. The example's rollout rule requires a positive 95% effect interval and less than 50 ms additional mean latency.

METHOD

12,000 generated users, one session each, approximately 50/50 random assignment. SQL and Python independently reconcile conversion counts. The analysis reports an unpooled 95% interval for the conversion-rate difference, a pooled two-sided z-test, and an assignment-balance check. Device cuts are exploratory.

RESULTS - SYNTHETIC DATA

Metric	Control	Treatment
Sessions	5983	6017
Conversion rate	10.43%	12.28%
Mean latency	479.0 ms	547.5 ms
No-result sessions	7.92%	5.75%

Conversion effect: **+1.85 percentage points**; 95% interval **[+0.72, +2.99]**. Two-sided p-value 0.0014. Mean latency increases by **68.5 ms**.

RECOMMENDATION

Hold the rollout under the stated guardrail. Conversion improves in this generated example, but the added latency exceeds 50 ms. Investigate the added search step, optimize it, and rerun a controlled experiment.

LIMITATIONS & REPRODUCIBILITY

The generator intentionally creates click-rate and latency differences. These are demonstration results, not discovered business impact. Real experiments also need prospective power planning, repeated-user handling, exposure validation, bot filtering, and seasonality checks. The fixed-seed Python script, SQL queries, notebook, and result files document the full calculation.

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